

SDS 315 Homework 8

Elizabeth 'Betsy' Sherhart

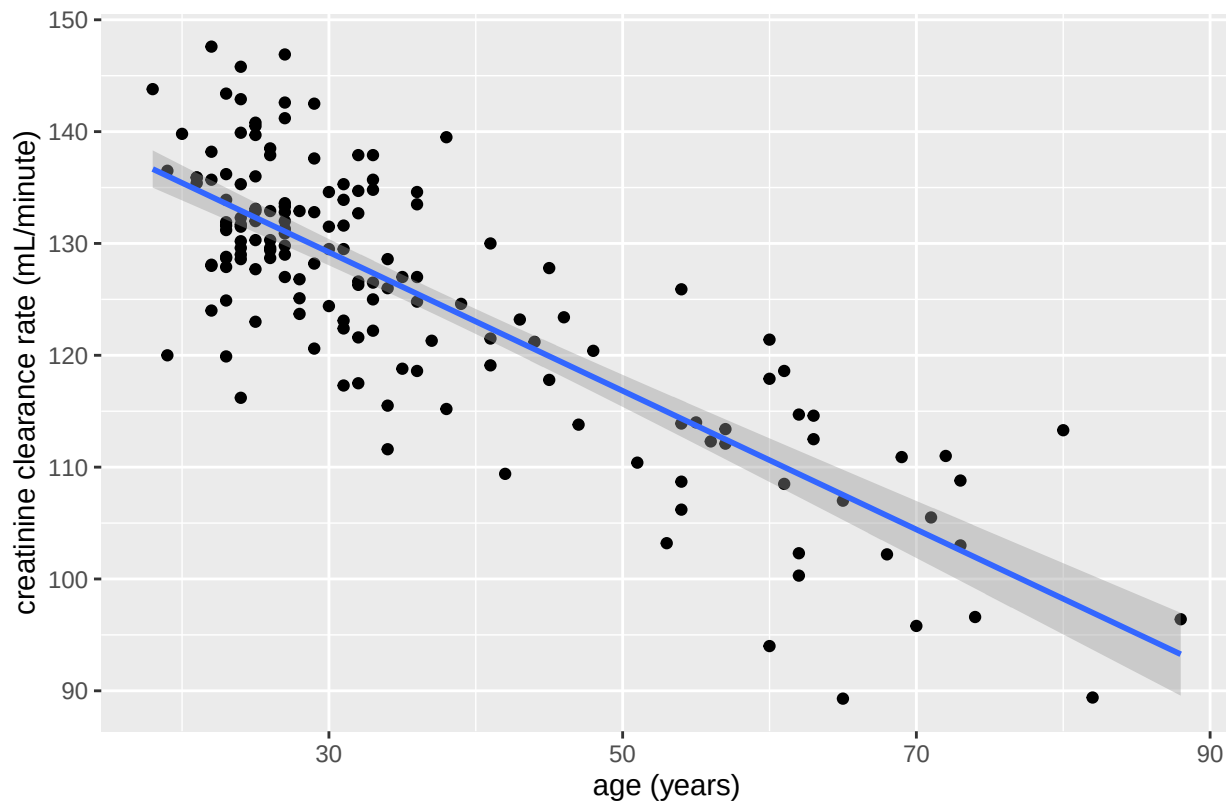
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April 14th, 2025

Problem 1: regression warm up

Creatinine clearance rate change with age



```
## (Intercept)      age  
## 147.8129158 -0.6198159
```

A) What creatinine clearance rate should we expect for a 55-year-old?

A 55-year-old is expected to have a creatinine clearance rate of 113.7 mL/minute based on the linear regression model.

This was calculated using the regression equation derived from the model:

$$\text{creatinineClearanceRate} = 147.8 - 0.62 * \text{age}$$

expected rate of 55 year old = $147.8 - 0.62 * 55 = 113.7$ mL/min

B) How does creatinine clearance rate change with age?

The creatinine clearance rate change with age is represented by the slope of the regression line, which is -0.62 mL/minute per year. This means that for every additional year of age, the creatinine clearance rate will decrease on average by 0.62 mL/minute.

C) Whose creatinine clearance rate is healthier (higher) for their age: a 40-year-old with a rate of 135, or a 60-year-old with a rate of 112?

A 40-year-old with a clearance rate of 135

A 60-year-old with a clearance rate of 112

Using the regression equation:

$$\text{creatinineClearanceRate} = 147.8 - 0.62 * \text{age}$$

Calculate the expected clearance rate for each and compare the residuals (actual - expected):

Calculate expected values: Expected for age 40:

$$147.8 - 0.62 \times 40 = 123$$

Residual:

$$135 - 123 = 12$$

Expected for age 60:

$$147.8 - 0.62 \times 60 = 110.6$$

Residual:

$$112 - 110.6 = 1.4$$

Since higher creatinine clearance rates indicate better kidney function, the 40-year-old has a healthier creatinine clearance rate for their age. Their clearance rate is 12 mL/min above the expected value, whereas the 60-year-old's clearance rate is only 1.4 mL/min above their expected value. Therefore, the 40-year-old exceeds their expected rate by a larger margin, indicating better kidney function relative to their age.

Problem 2: Modeling disease growth

1. Estimated growth rate and doubling time for Italy, with 95% bootstrapped confidence intervals for each

Italy

```
##               name      lower    upper level    method estimate
## 2 days_since_first_death 0.1588725 0.2083307   0.95 percentile 0.183218
```

By using a linear model I was able to understand how daily COVID-19 deaths increased in Italy over time. The estimated growth rate for Italy was about 0.183, indicating that the deaths increased by 18.3 % every day. The estimated doubling time was 3.8 days. The 95% bootstrapped confidence intervals for the growth rate ranged from around 0.159 to 0.208. The 95% bootstrapped confidence intervals for the doubling time ranged from around 4.4 days and 3.4 days.

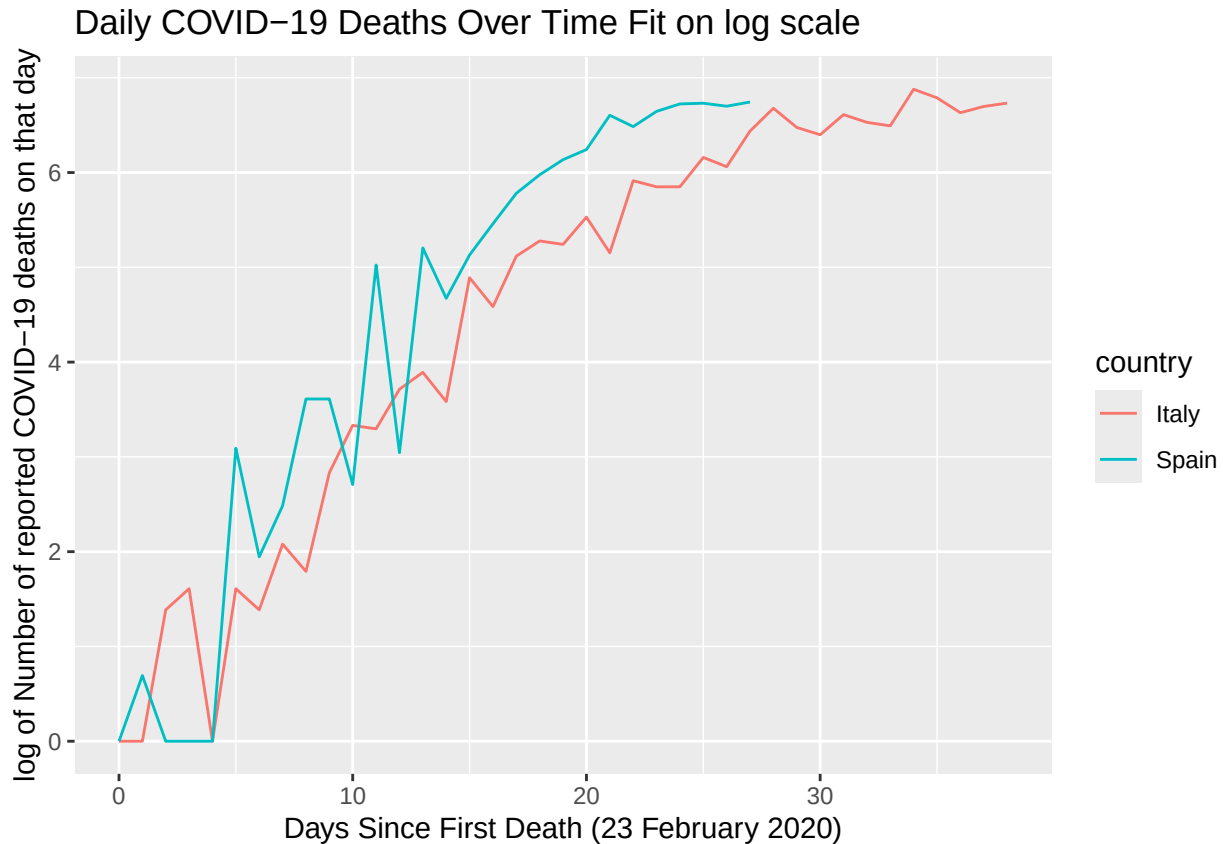
2. Estimated growth rate and doubling time for Spain, with 95% bootstrapped confidence intervals for each

Spain

```
##               name      lower    upper level    method estimate
## 2 days_since_first_death 0.2350455 0.3176832   0.95 percentile 0.2762447
```

By using a linear model I was able to understand how daily COVID-19 deaths increased in Spain over time. The estimated growth rate for Spain was 0.276, indicating that the deaths increased by 27.6% every day. The estimated doubling time was 2.5 days. The 95% bootstrapped confidence intervals for the growth rate ranged from around 0.235 and 0.318. The 95% bootstrapped confidence intervals for the doubling time ranged from around 3.0 days and 2.2 days.

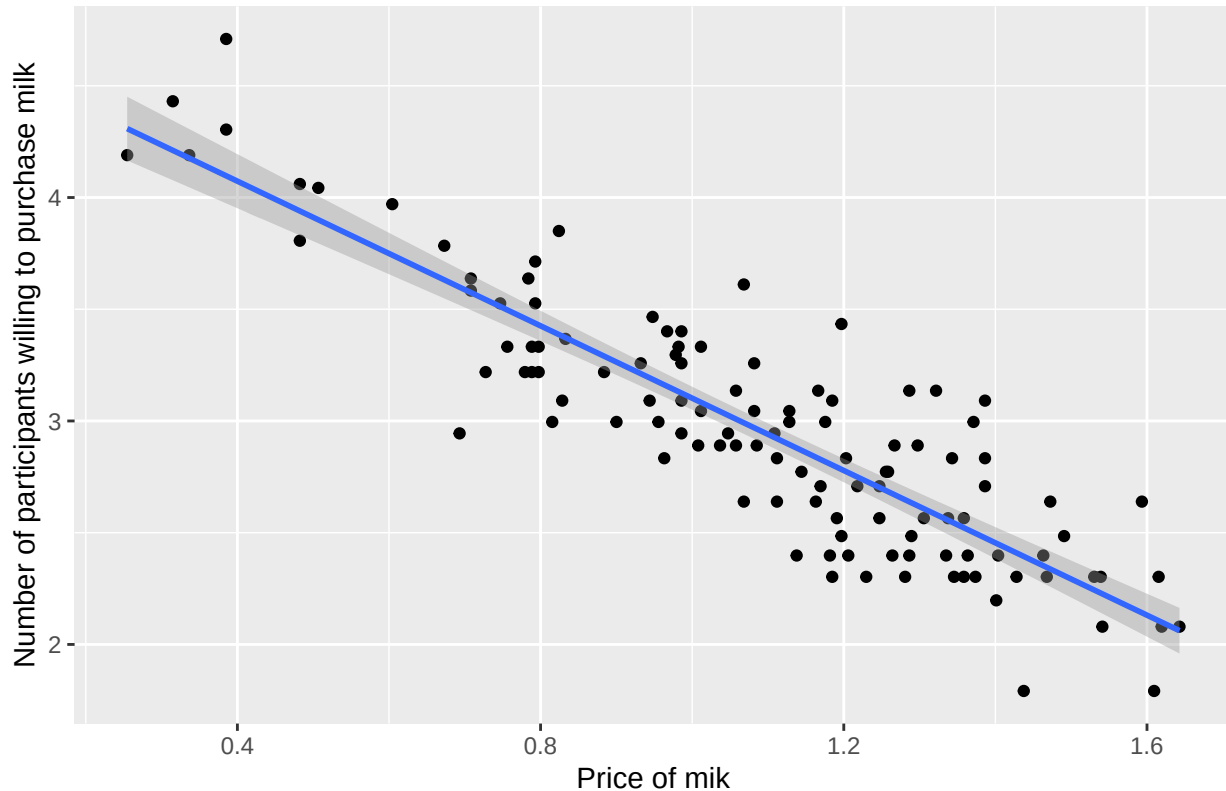
3.



The graph represents the daily COVID deaths that happened over time starting from February 23rd, 2020 in Italy and Spain. The graph provides a visual comparison of the rate of increasing deaths between Italy and Spain. Spain has a more steep slope than Italy indicating that Spain had a higher growth rate than Italy. This is supported by Spain's greater growth rate and shorter doubling time as calculated earlier.

Problem 3: price elasticity of demand

Price elasticity of demand for milk



```
##      name      lower      upper level      method estimate
## 2 log_price -1.773498 -1.451893  0.95 percentile -1.618578
```

The estimated price elasticity of demand for milk is -1.62, with a 95% bootstrapped confidence interval ranging from around -1.77 to -1.46.

To estimate this value, I used a power-law model of the form:

$$Q = K \cdot P^\beta$$

By taking the logarithm of both sides, I transformed it into a linear model:

$$\log(Q) = \log(K) + \beta \cdot \log(P)$$

I fit this transformed model using linear regression of $\log(\text{sales})$ on $\log(\text{price})$ and bootstrapped the model 10,000 times to obtain a confidence interval for the elasticity estimate β . Since the slope of this regression corresponds to the price elasticity, a value of -1.62 implies that a 1% increase in price leads to a 1.62% decrease in milk sales, indicating that demand is elastic. Additionally, because the confidence interval does not include 0, the result is statistically significant and confirms that price of milk has a meaningful impact on its demand.